Hugo Marquerie 28/04/2026

Lem Convergencia Uniforme Compactos Imp Continuidad

Lema 1. *Sea* $\Omega \subset \mathbb{C}$ *y sean* $(f_n)_{n\in\mathbb{N}} \subset \mathcal{C}(\Omega)$ *y* $f\colon \Omega \to \mathbb{C}$, *entonces*

$$f_n \xrightarrow[n\to\infty]{\Omega} f \implies f \in \mathcal{C}(\Omega).$$

***Demostración*:** Sea $a \in \Omega$ y $r > 0$ tal que $D(a,r) \subset \overline{D(a,r)} \Subset \Omega$, entonces $f_n \xrightarrow[n\to\infty]{\overline{D(a,r)}} f$. Sea $\varepsilon > 0$, entonces $\exists N \in \mathbb{N} : \forall n \geqslant N : \forall z \in \overline{D(a,r)} : |f_n(z) - f(z)| < \varepsilon/3$. Tomamos dicho $N \in \mathbb{N}$ y $\delta \in (0,r)$ tal que $\forall z \in D(a,\delta) : |f_N(z) - f_N(a)| < \varepsilon/3$ por continuidad de f_N, entonces $\forall z \in D(a,\delta)$:

$$\begin{aligned}|f(z) - f(a)| &= |f(z) - f_N(z) + f_N(z) - f_N(a) + f_N(a) - f(a)| \\ &\leqslant |f(z) - f_N(z)| + |f_N(z) - f_N(a)| + |f_N(a) - f(a)| < 3 \cdot \varepsilon/3 = \varepsilon.\end{aligned}$$ ■

Referenciado en

- Lem convergencia uniforme compactos imp convergencia sucesion
- Lem Convergencia Uniforme Compactos Subsubsucesiones

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